Problem Set 4B

Resolution of a vector into rectangular components

Do these after the slides for Section 4B. Textbook Exercise 4B: Questions 9, 10, 11, 12, 20, 31. Answers are on the last page. Show full working in your exercise book.

Skills

1 Let $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find:

(a) $\mathbf{a} + 2\mathbf{b}$

(b) $3\mathbf{a} - \mathbf{b}$

(c) $|\mathbf{a}|$

(d) $|\mathbf{a} - \mathbf{b}|$

2 For the points $A(3, -1, 2)$ and $B(-1, 1, 6)$, find:

(a) $\overrightarrow{AB}$

(b) $|\overrightarrow{AB}|$

(c) the unit vector in the direction of $\overrightarrow{AB}$

(d) the coordinates of the midpoint of AB

3 Find the vector of magnitude 14 in the direction of $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.

4 The vectors $3\mathbf{i} + p\mathbf{j} - 6\mathbf{k}$ and $q\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ are parallel. Find p and q.

5 Find, correct to one decimal place, the angle that $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ makes with:

(a) the positive x-axis

(b) the positive y-axis

Exam style

6 Let $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{c} = \mathbf{i} + 7\mathbf{j} + t\mathbf{k}$. Find t if $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are linearly dependent.

7 $ABCD$ is a parallelogram with $A(2, 1, -1)$, $B(5, 3, 0)$ and $D(1, 4, 2)$. Find the coordinates of C and the lengths of the two diagonals.

8 $A(1, 2, 3)$ and $B(4, -2, 3)$. Find the coordinates of the point P on the segment AB that is 2 units from A.

Problem Set 4B: answers

1 (a) $\mathbf{j} + 9\mathbf{k}$ (b) $7\mathbf{i} - 11\mathbf{j} - \mathbf{k}$ (c) $\sqrt{14}$ (d) $\sqrt{43}$

2 (a) $-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ (b) 6 (c) $\frac{1}{3}(-2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ (d) $(1, 0, 4)$

3 $4\mathbf{i} - 6\mathbf{j} + 12\mathbf{k}$

4 $p = -12,\ q = -1$

5 (a) $48.2°$ (b) $109.5°$

6 $\mathbf{c} = 3\mathbf{a} - \mathbf{b}$, so $t = -6$

7 $C(4, 6, 3)$; $AC = 3\sqrt{5}$, $BD = \sqrt{21}$

8 $|\overrightarrow{AB}| = 5$, so $P = \left(\frac{11}{5}, \frac{2}{5}, 3\right)$